Exploratory Data Analysis - (EDA) & Insights Report

SALES & CUSTOMER BEHAVIOR ANALYSIS REPORT

Prepared for: Mall Management Team

Date: 4/11/2026

Mall Revenue Optimization Through Customer Behavior Insights

Executive Summary

This report analyzes a retail dataset including 99,457 transactions with customer demographics and sales details. The purpose of analysis is to get useful insights which can be helpful for management of malls to generate more profits.

Dataset Overview:

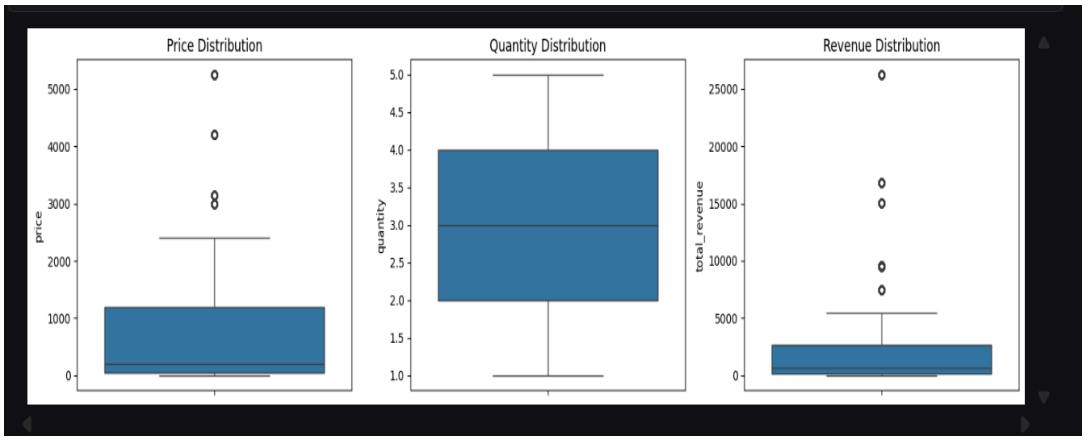
- Source: Kaggle ("Sales and Customer data")
- Period: March 2021 - October 2022 (based on invoice dates)
- Size: 99,457 transactions, linking 99,457 unique customer records.
- Files:
    - **customer_data.csv: Demographics (Customer ID, Gender, Age, Payment Method)**
    - **sales_data.csv: Transaction details (Invoice No, Customer ID, Category, Quantity, Price, Date, Mall)**

Data Cleaning & Preparation:

- Missing Values: Missing ages (total 119) were imputed with median ages as there could be some outliers.
- Data Types: Types: Type of "invoice_date" feature was successfully changed from string to datetime.
- Duplicates: There were no duplicates found in both datasets.
- Feature Engineering: Added "total_revenue" as the multiplication of "price" and "quantity". Added age groups (Young, Middle, Senior).

2. Visual Analysis

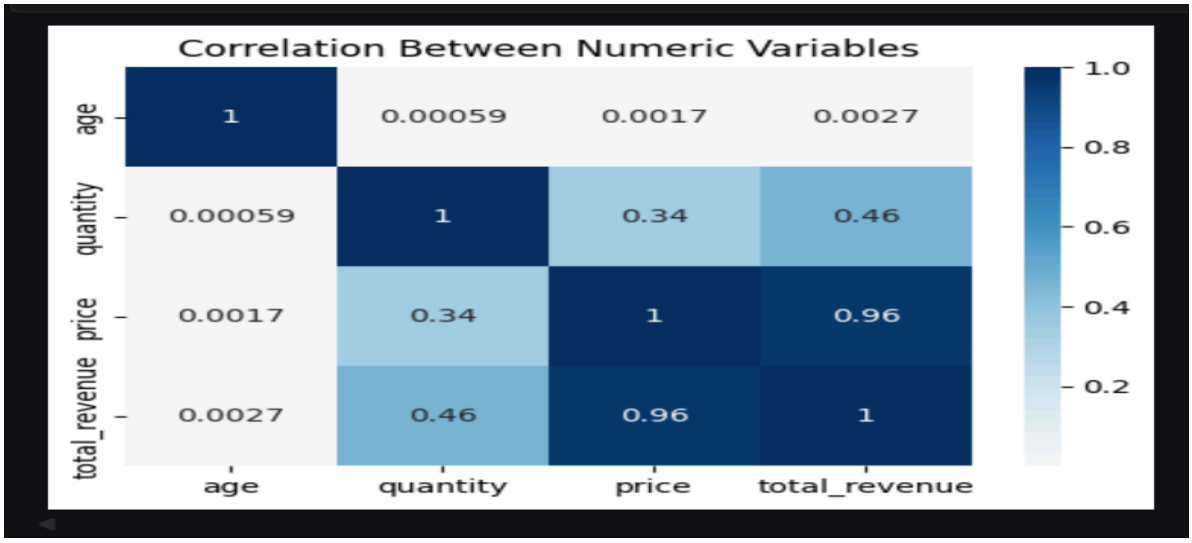
2.1 Boxplot Analysis - Outlier Detection



1. Distribution and Outliers

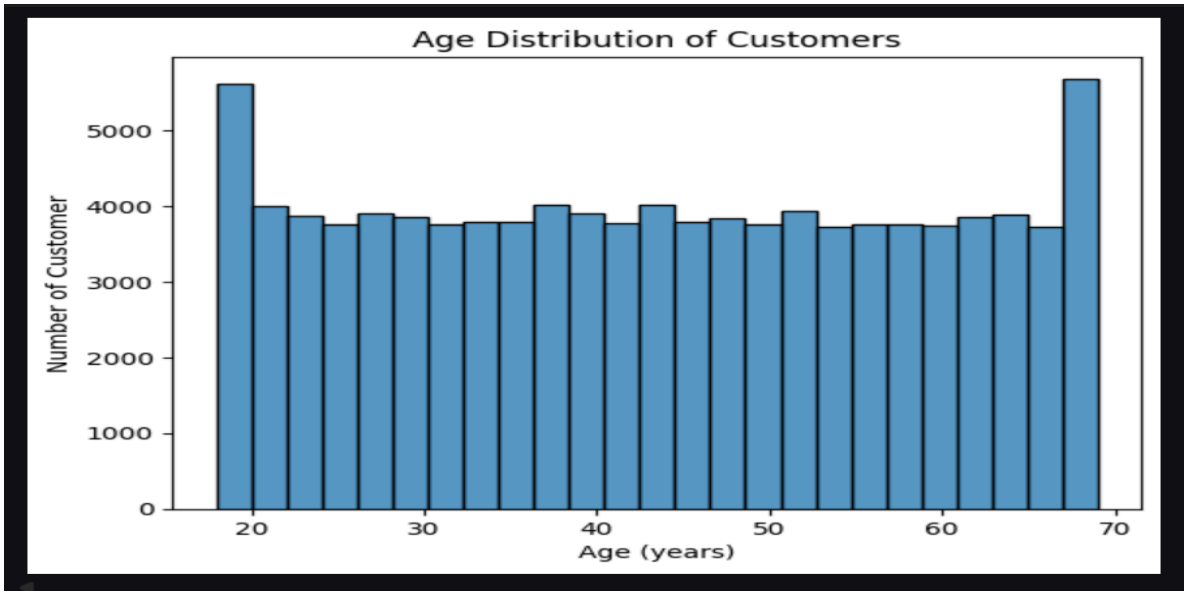
- Price and Total Revenue Boxplots: The boxplot successfully found major outliers for price and total_revenue. Around 2.27% of sales are considered price outliers, which means they involve expensive luxury goods (such as Technology products worth $4,200). These outliers are important to revenue and must remain.
- Quantity Boxplot: Most purchases include between one and five units, although some larger orders are occasionally placed.

2.2 Correlation Heatmap



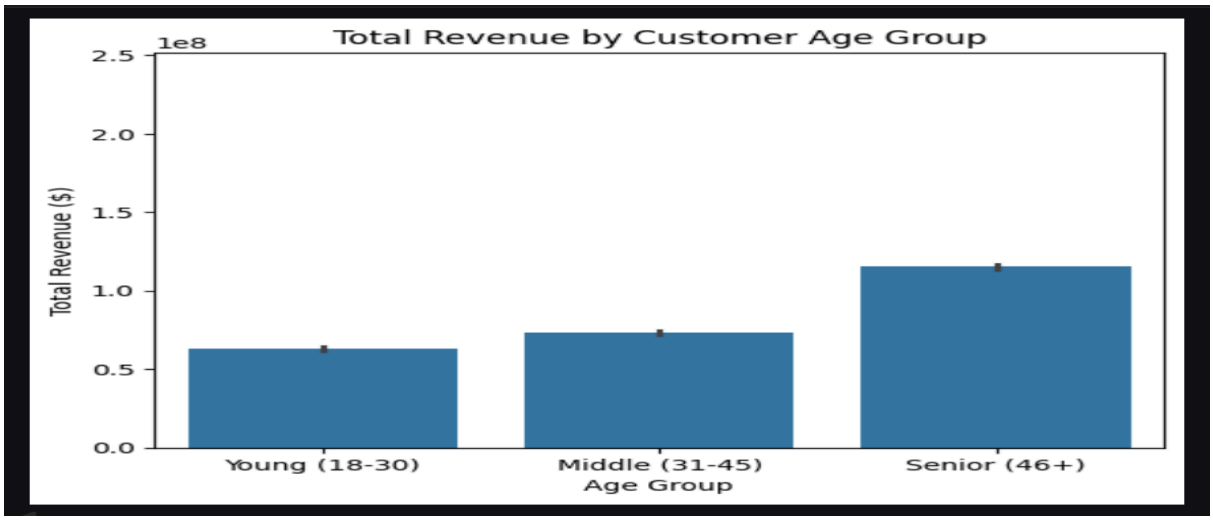
- total_revenue is significantly correlated with the price variable but not with the quantity variable. This means that higher revenue levels result from the sale of costly products rather than large quantities.
- Age is uncorrelated with any other numeric variables.

2.3 Age Distribution



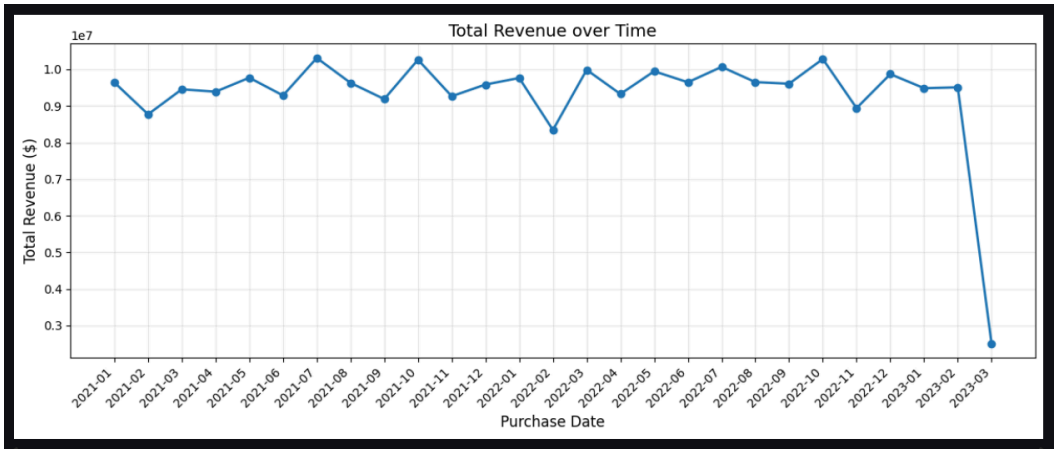
- Histogram: There is an extensive range of customers in terms of their ages with a significant mode in the 30-40 age group.
- Bar Graph for Revenue by Age Groups: The elderly customers (46+) generate the highest revenues followed by the middle customers (31-45) and young customers (18-30)

2.4 Revenue by Age Group



- The line plot reveals seasonality. Revenue peaks around January and dips in November, suggesting a strong holiday/New Year spending pattern.

2.5 Monthly Revenue Trend



- Your line plot effectively reveals seasonality. Revenue peaks are visible around January and dips in November. This suggests a strong holiday/New Year spending pattern.

3. Ten Key Insights & Recommendations

Insight 1: Seniors spend the most- young people spend the least.

Recommendation: Organize a special “Senior Citizen’s Day” where you give discounts and additional loyalty points. Young customers will be offered student discounts and low-cost beginner products.

Insight 2: The young generation generates the smallest amount of revenues.

Recommendation: Organize some fun social media challenges and pop-ups that appeal to the tastes of students and young adults. Partner with student discount mobile applications and make their presence more enticing at the shopping complex

Insight 3 and 4: More women than men come to the malls, and they spend the most money.

Recommendation: Market hard to the female demographic, and provide entertainment features such as fashion seminars and well-being events. Ensure that facilities such as comfortable toilets and lounges cater to women.

Insight 5: Customers mainly use cash, followed by card payments.

Recommendation: Instead of eliminating cash, provide a 5% discount on transactions using cards to encourage consumers to spend more. Issue your own card, which rewards customers for their purchases.

Insight 7: Mall of Istanbul is rocking in comparison to other sites.

Recommendation: Identify and implement in other malls such as Kanyon and Metrocity the factors that make Mall of Istanbul perform so well.

Insights 6 and 8 : People love purchasing clothes while souvenirs generate little interest.

Recommendation: Make clothing sales into events such as a "Fall Fashion Festival" and bundle up souvenirs for half price if customers buy clothing.

Insight 9: C100004 is the single most frequent buyer of all your customers.

Recommendation: Create a tiered VIP loyalty program with Platinum perks like a personal shopper. Customer C100004 shows high purchase frequency and loyalty potential.

Insight 10: January is Peak Month

Recommendation: A Winter Clearance sale can be organized from the first to the second week of January. January is the peak revenue month and requires strong promotional activities.

By implementing the recommendations above, the mall management can expect to see measurable revenue growth within 3-6 months. Regular monitoring of monthly revenue trends and segment performance is advised to adjust strategies as needed.